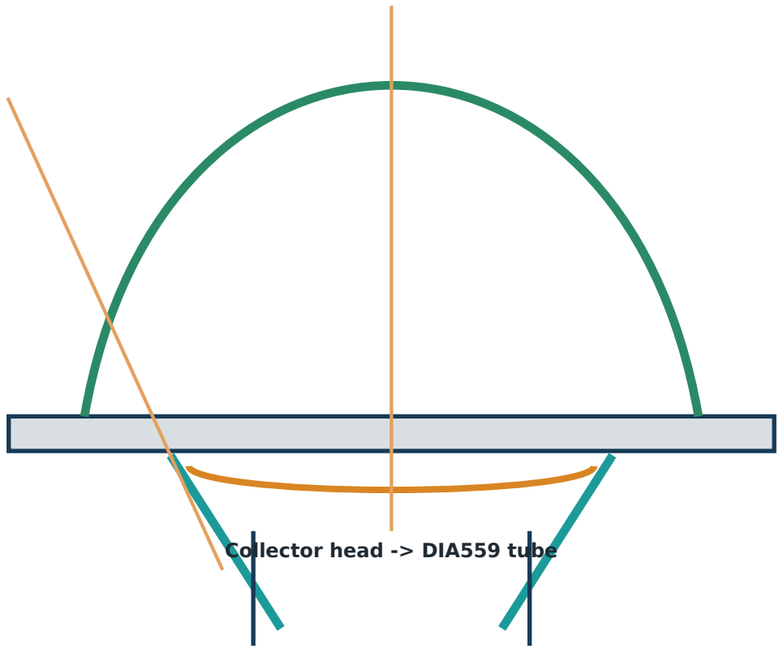
DOME SUNLIGHT COLLECTOR - ENHANCED FABRICATION GUIDE

Readable companion to the 5-sheet prototype collector-head package



Purpose

This edition preserves the original prototype collector geometry and test requirements while adding a clearer component map, forming sequence, optical explanation, assembly logic, and release checklist. It is designed to help a fabricator or reviewer understand what is fixed for the prototype and what still requires project engineering.

ENGINEERING HOLD - PERMANENT ROOF INSTALLATION

The source package does not define the final roof curb, flashing, anchors, or structural connection. Those items require roof type/slope, project location, adopted code, wind pressure, snow load, and related classifications.

559 mm tube clear diameter	**650 mm** dome forming aperture	**300 mm** dome rise
3.2 mm minimum formed PC wall	**95%+** redirector reflectance	**60 deg** redirector surface angle

How to use this file

- Original sheets are preserved as the dimensional source.
- Explanation pages follow each sheet and translate the drawing into fabrication actions and checks.
- HOLD items remain HOLD. This explanatory edition does not create a building-installation release.
- Prototype tests must be documented before any project-specific installation set can be issued.

COLLECTOR QUICK REFERENCE

Component stack, critical geometry, and project data still required

COMPONENT STACK - TOP TO BOTTOM

D-01	**UV-stabilized polycarbonate dome** 8.0 mm starting sheet; formed wall 3.2 mm minimum.
C-01	**304 SS clamp ring** OD715 / ID652 x 4; clamps the dome while allowing thermal sliding.
S-01 / S-02	**EPDM seals** 6 mm closed-cell flange gasket plus 6 mm O-ring cord.
B-01	**304 SS base plate** OD780 x 3 with DIA559 tube aperture.
G-01	**Condensate gutter** OD640 / ID560 x 18 deep; four drains to exterior.
O-01	**Specular aluminum redirector** DIA600 top / DIA400 clear opening x H173.2; 60 deg mirror angle.

FIXED PROTOTYPE GEOMETRY

- Tube clear diameter: 559 mm (22 in nominal).
- Dome: 650 mm forming aperture x 300 mm rise; spherical radius R326.
- Dome material: UV polycarbonate, 8.0 mm starting sheet, 3.2 mm minimum after forming.
- Base: 304 stainless steel, OD780 x 3, aperture DIA559.
- Redirector: DIA600 / DIA400 x H173.2, beta = 60 deg.
- Seal concept: 6 mm EPDM gasket + secondary 6 mm O-ring.

PROJECT DATA REQUIRED BEFORE ROOF RELEASE

- Roof system / material / slope
- Project location + adopted code edition
- Positive and negative design wind pressure
- Ground and roof snow loads
- Hail / impact classification
- Fire / roof-covering classification
- Curb height, insulation and flashing system

ROOF HOLES ARE A HOLD ITEM

The PCD750 roof/curb hole pattern is provisional. Do not drill those holes until curb geometry, anchor edge distance, wind uplift, and flashing are approved.

OPTICAL INTENT

The dome accepts diffuse and direct light. The annular 60 deg redirector is tuned so a 30 deg solar-altitude ray is redirected approximately vertical; high-angle sunlight can pass through the DIA400 central opening.

DOME SUNLIGHT COLLECTOR - GENERAL ARRANGEMENT

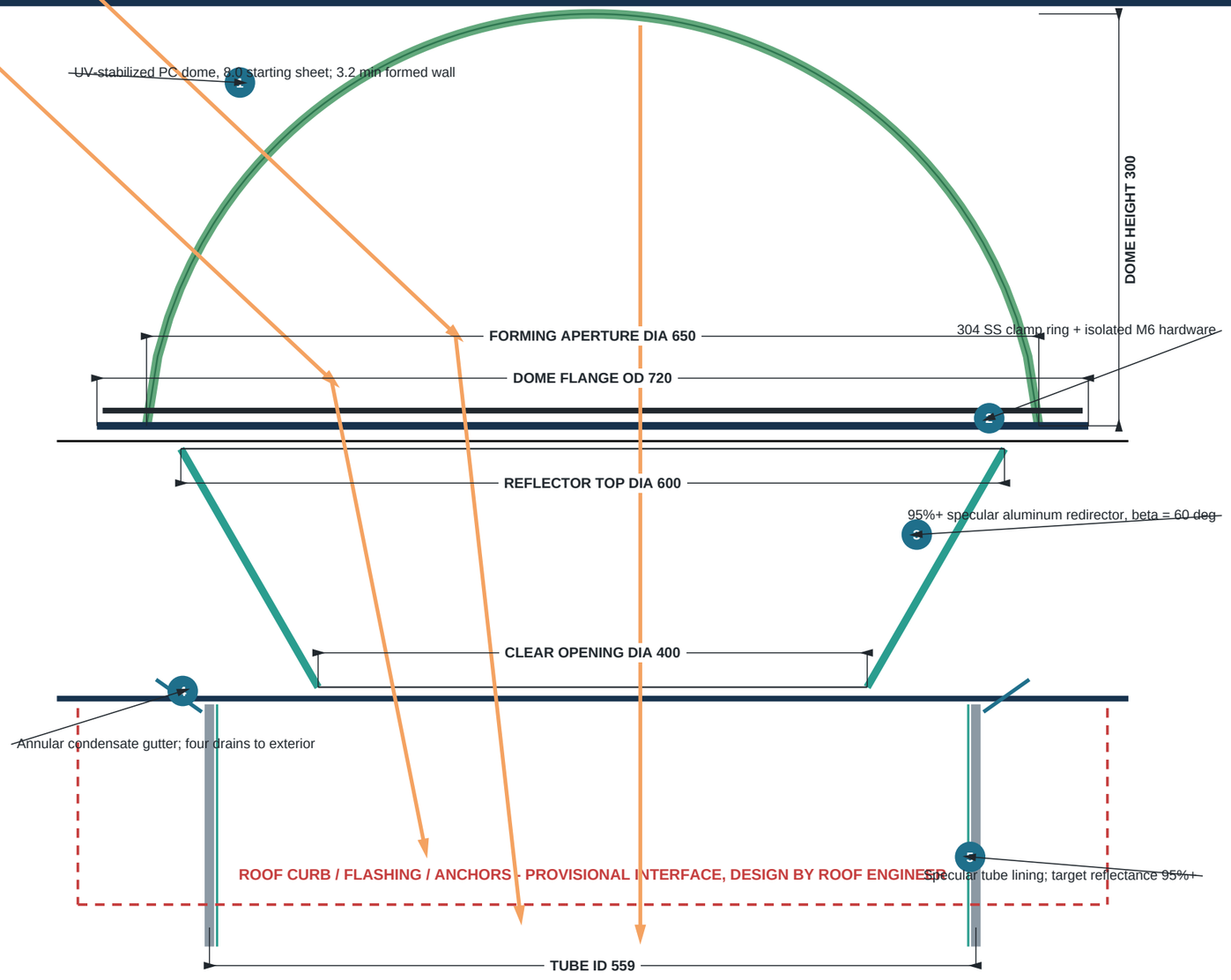
Roof-interface-neutral prototype; non-proprietary optical collector head for a nominal 559 mm tubular daylight device

UNITS: mm unless noted | SCALE: diagrammatic

ENGINEERING HOLD POINT - PERMANENT ROOF INSTALLATION IS NOT YET DEFINED

Build the collector head as a prototype only. Final curb, flashing, anchors and structure require roof type/slope, project location, adopted code, wind pressure and snow load.

SECTION A-A - COLLECTOR HEAD AND CONTROLLED ROOF INTERFACE



DESIGN BASIS - FIXED FOR PROTOTYPE

Tube clear diameter	559 (22 in nominal)
Dome geometry	650 aperture x 300 rise; spherical R326
Dome material	UV PC, 8.0 starting; 3.2 min formed
Collector base	304 SS, OD780 x 3; aperture DIA559
Optical redirector	DIA600 / DIA400 x H173.2; 60 deg
Seal concept	6 EPDM gasket + secondary 6 O-ring
Roof interface	PCD750 provision only - NOT RELEASED

PROJECT INPUTS REQUIRED BEFORE ROOF RELEASE

Roof system / material / slope	TBD
Project location + adopted code edition	TBD
Design wind pressure: positive and negative	TBD
Ground and roof snow loads	TBD
Hail / impact classification	TBD
Fire / roof-covering classification	TBD
Curb height, insulation and flashing system	TBD

OPTICAL PRINCIPLE AND PERFORMANCE INTENT

1. The clear dome accepts diffuse sky light and direct sun over a wide azimuth range.
2. The annular 60 deg mirror redirects a 30 deg solar-altitude ray approximately vertical.
3. The DIA400 central opening passes high-angle sunlight directly into the tube.
4. For solar altitude alpha, reflected deviation from vertical is approximately alpha - 30 deg.
5. Optical output is a design intent only; verify with lux mapping at 15, 30 and 60 deg incidence.

SHEET 1 EXPLAINED - GENERAL ARRANGEMENT

What is fixed for the prototype and what is intentionally left to the roof engineer

OPTICAL PATH IN PLAIN LANGUAGE

- Diffuse sky + direct sun enter through the clear dome.
- Low-angle sunlight can strike the annular specular redirector and be turned toward the tube axis.
- High-angle sunlight can pass through the DIA400 central opening without using the redirector.
- For the 2D design relation, reflected deviation from vertical is approximately alpha - 30 deg.
- Final performance must be verified by lux mapping at 15, 30 and 60 deg incidence, not by geometry alone.

CONTROLLED VS UNCONTROLLED INTERFACES

Controlled in this prototype package:

- Dome and base geometry
- Tube aperture DIA559
- Redirector geometry and reflectance intent
- Seal concept
- Condensate gutter concept

Still project-specific:

- Roof curb and flashing
- Anchors and structure
- Roof loads and code edition
- Hail/fire classifications
- Curb insulation and final weatherproofing

RELEASE CHECK - DO NOT SUBSTITUTE ASSUMPTIONS

Roof system / material / slope TBD

Location + code edition TBD

Wind pressure (+/-) TBD

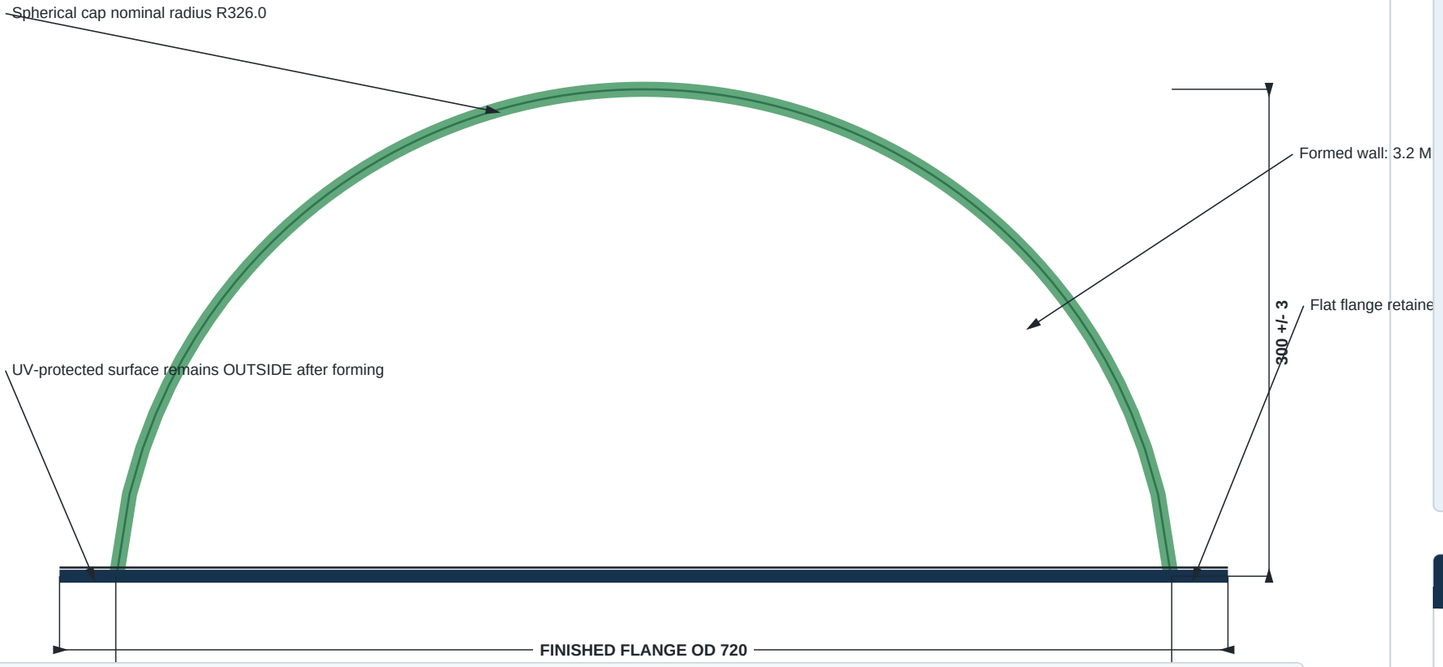
Ground / roof snow loads TBD

Hail / impact classification TBD

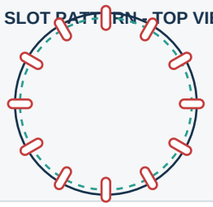
Fire / roof-covering classification TBD

Curb height + flashing system TBD

PART D-01 - FORMED POLYCARBONATE DOME (SECTION)



FLANGE SLOT PATTERN - TOP VIEW (schematic)



12 radial slots 8 x 18 equally spaced at 30 deg on PCD680

Slot long axis radial; deburr and polish edges; no sharp internal corners

Slots accommodate about 1.9 mm radial growth for Delta-T = 80 C

FORMED-WALL MATERIAL BUDGET - WHY THE STARTING SHEET IS 8 mm

Spherical cap area = 2 x pi x R x h = 2 x pi x 326.0 x 300 = 614,500 mm2 (nominal)

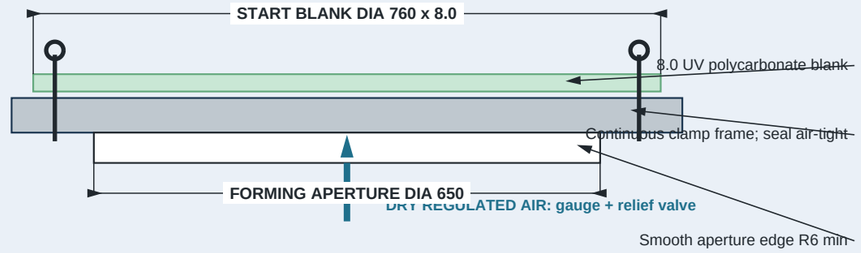
Forming aperture area = pi x 325 x 325 = 331,800 mm2 | draw ratio = 614,500 / 331,800 = 1.852

Ideal average cap wall from 8.0 sheet = 8.0 / 1.852 = 4.32 mm before local thinning and process variation.

A 5.0 mm sheet would average only 2.70 mm, so it cannot reliably meet a 3.2 mm minimum formed wall.

This calculation does not replace thickness mapping. Reject any part with a reading below 3.2 mm.

STARTING BLANK AND FREE-BLOW FIXTURE



CONTROL	PROTOTYPE SETTING / ACCEPTANCE
Material	UV-stabilized optical PC; protected both sides
Pre-dry 8 mm	125 +/- 3 C for 36 h; follow sheet supplier
Forming window	170-180 C suggested for dome forming trials
Pressure	Trial-determined; regulator, gauge and relief required
Height stop	300 +/- 3; calibrated non-contact gauge
Release	Hold shape until rigid; about 120-125 C part temp

DOME QA, TRIM AND HANDLING

1. Mark a 50 x 50 inspection grid over the cap.
2. Map ultrasonic or mechanical thickness at every node.
3. Accept only if every cap reading is 3.2 mm or greater.
4. Trim finished flange to DIA720 concentric within 1.0 TIR.
5. Machine 12 slots 8 x 18 on PCD680; edge radius 1 min.
6. Inspect for haze, bubbles, webbing, scratches and UV-side reversal.
7. Do not flame-polish PC; avoid stress-cracking cleaners/sealants.
8. Record sheet lot, heat cycle, height, pressure history and thickness map.

PROCESS SAFETY

Heated sheet and compressed air store significant energy. Fixture, pressure limits, guarding and operator procedures must be engineered by the forming shop.

SHEET 2 EXPLAINED - DOME FORMING

Material budget, forming controls, and the evidence required for acceptance

FORMING SEQUENCE

1. 8 mm UV-stabilized optical polycarbonate; source prototype setting 125 +/- 3 C for 36 h, following supplier guidance.
2. **Heat**
   Suggested forming-trial window 170-180 C.
3. **Clamp + seal**
   Use continuous frame over a DIA650 forming aperture with smooth R6 minimum edge.
4. **Free-blow**
   Use dry, regulated air with gauge and relief valve; pressure is trial-determined.
5. **Stop at height**
   Control dome rise to 300 +/- 3 using a calibrated non-contact gauge.
6. **Hold until rigid**
   Source gives about 120-125 C part temperature as prototype release point.
7. **Map thickness**
   50 x 50 mm inspection grid; reject any cap reading below 3.2 mm.
8. **Trim + slot**
   Trim flange to DIA720 and machine twelve 8 x 18 radial slots on PCD680.

WHY THE STARTING SHEET IS 8 mm

The source uses a geometric material-budget check for the spherical cap:

Cap area = 614,500 mm2

Aperture area = 331,800 mm2

Draw ratio = 1.852

8.0 / 1.852 = 4.32 mm avg.

This average does not guarantee local thickness. The release criterion is still the actual thickness map: every measured cap point must be at least 3.2 mm.

DOME QA RECORD - CAPTURE THESE VALUES

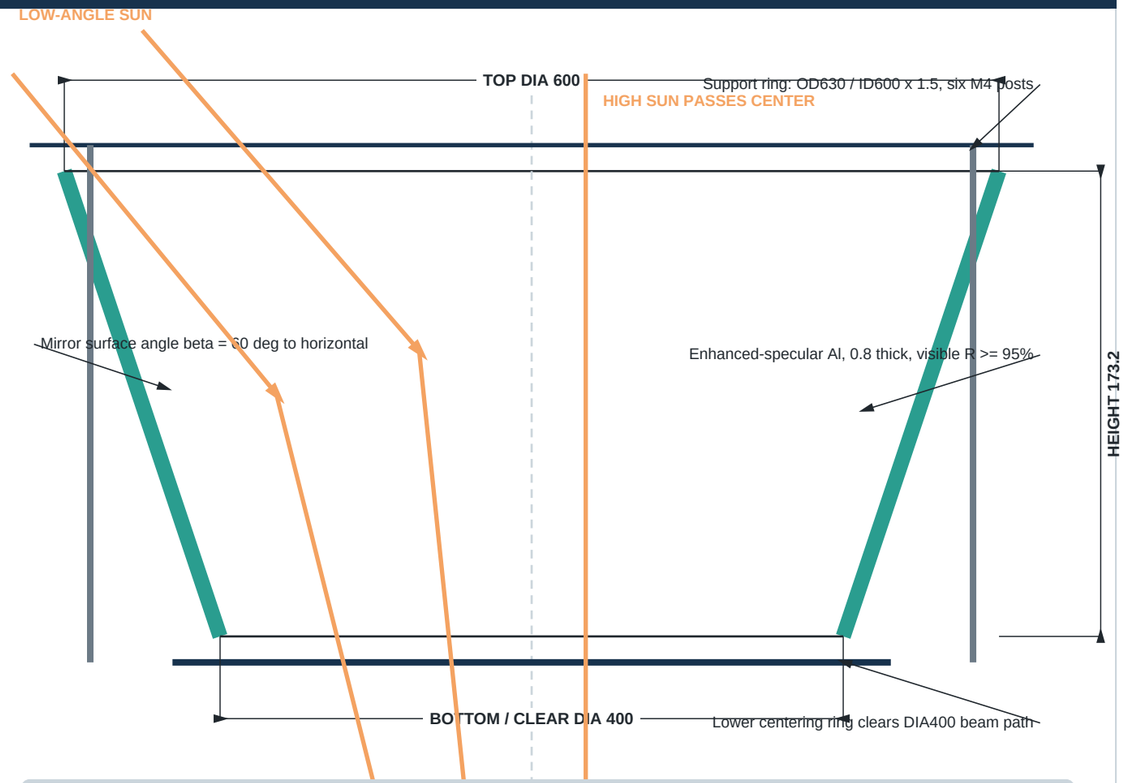
Material traceability Sheet lot / UV side	**Thermal process** Pre-dry + heat cycle	**Pressure process** Pressure history	**Geometry** Height + flange concentricity
Thickness 50 mm grid map	**Visual** Haze / bubbles / webbing / scratches	**Slots** 12 x 8x18 on PCD680	**Handling** No flame polish; approved cleaners/sealants

OPTICAL REDIRECTOR - CONE, FLAT PATTERN AND RAY DESIGN

Annular enhanced-specular aluminum insert; generic geometry optimized for 30 deg solar altitude

UNITS: mm unless noted | SCALE: diagrammatic

PART O-01 - ANNULAR CONICAL REFLECTOR

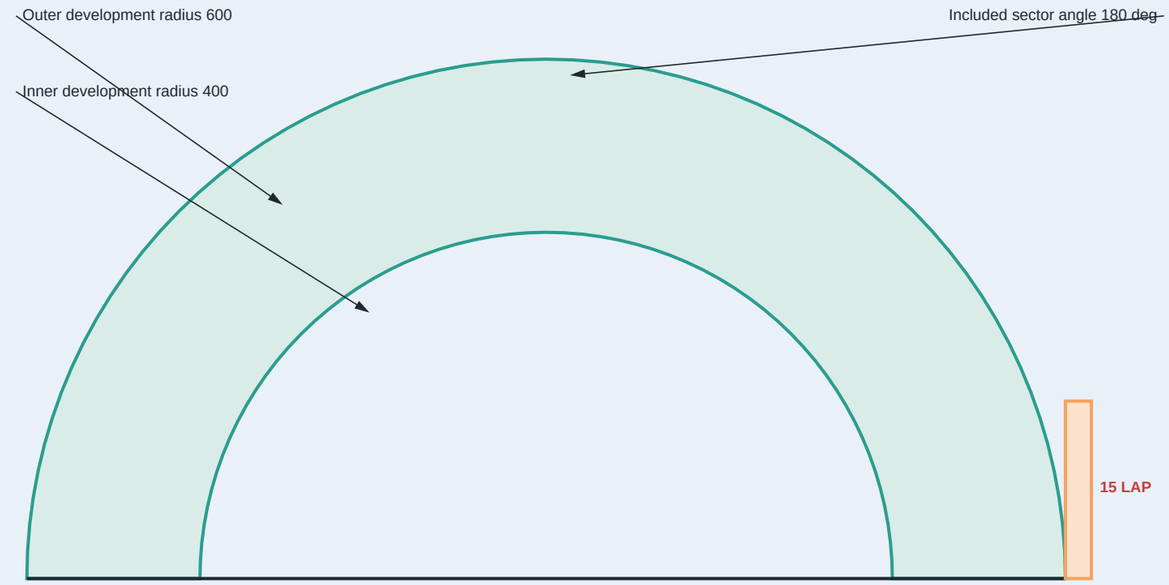


Finished slant length: 200.0 | radial change: 100.0 | axial height: sqrt(200^2 - 100^2) = 173.2

Roll with protective film on optical face. Seam at north/reference side; 15 lap, flush rivet or bonded lap.

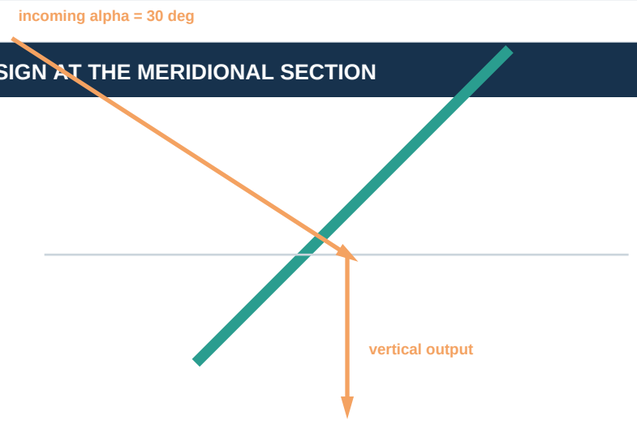
No fastener head may protrude into the mirror face or the DIA400 central clear opening.

FLAT PATTERN O-01 - HALF-ANNULAR SECTOR



CUT SIZE derives from cone radii 300/200 and slant length 200. Verify a paper template before cutting optical aluminum.

RAY DESIGN AT THE MERIDIONAL SECTION



SPECULAR RELATION

beta = 45 deg + alpha0 / 2 = 60 deg for alpha0 = 30 deg

Approximate reflected deviation from vertical: delta = alpha - alpha0

alpha = 15 deg -> delta = -15 deg | alpha = 30 deg -> 0 deg | alpha = 60 deg -> +30 deg

This 2D law sets the cone slope. Full 3D performance varies with solar azimuth, dome refraction, surface errors and tube losses.

Acceptance is photometric: lux map at the diffuser for 15, 30 and 60 deg elevation and multiple azimuths.

REDIRECTOR FABRICATION CONTROLS

1. Optical-face reflectance certificate: 95% visible or better.
2. Cut half-annulus R600/R400, 180 deg; add 15 lap.
3. Roll without scratching; use soft covered rolls and gloves.
4. Join seam behind lap; keep inner face smooth and clean.
5. Finished DIA600 / DIA400 / H173.2, each +/- 1.0.
6. Total runout at support ring: 1.5 max.
7. Clean only with material-approved non-abrasive method.
8. Orient seam to roof reference mark for repeatable testing.

SHEET 3 EXPLAINED - OPTICAL REDIRECTOR

Cone development, ray-design logic, and shop controls for the specular surface

O-01 FINISHED GEOMETRY

- Top diameter: 600 mm.
- Bottom / clear diameter: 400 mm.
- Height: 173.2 mm.
- Slant length: 200.0 mm.
- Mirror angle: beta = 60 deg to horizontal.
- Material: enhanced-specular aluminum, 0.8 mm, visible reflectance >=95%.
- Support: OD630 / ID600 x 1.5 ring with six M4 posts.

FLAT PATTERN - WHAT TO CUT

The conical reflector is developed as a half-annular sector:

R600	R400	180 deg
outer development radius	inner development radius	included sector angle

SEAM ALLOWANCE

Add a 15 mm lap. The source requires a smooth optical face and no fastener head in the mirror face or DIA400 clear opening.

RAY-DESIGN LOGIC

For the 2D meridional design, the source uses:

beta = 45 deg + alpha0 / 2

alpha0 = 30 deg -> beta = 60 deg

Approximate reflected deviation from vertical is delta = alpha - alpha0. Therefore 15 deg input gives -15 deg, 30 deg gives 0 deg, and 60 deg gives +30 deg. The source warns that real 3D performance also depends on azimuth, dome refraction, surface errors, and tube losses.

FABRICATION CONTROLS

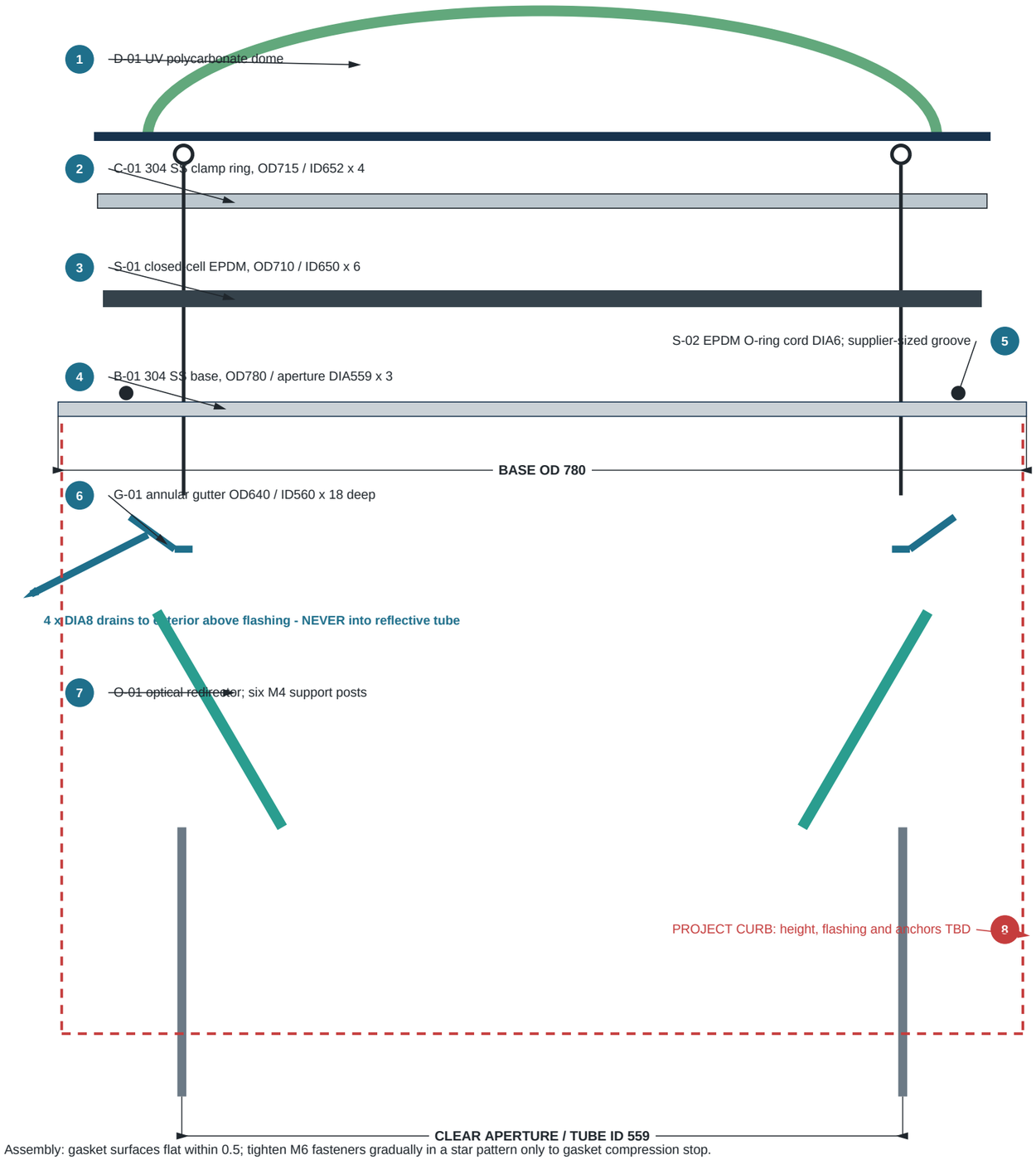
- Reflectance certificate >=95% visible.
- Roll with protective film, soft covered rolls and gloves.
- Keep the inner mirror face smooth and clean.
- Finished DIA600 / DIA400 / H173.2 each +/- 1.0.
- Total runout at support ring <= 1.5 mm.
- Orient seam to roof reference mark for repeatable testing.
- Acceptance remains photometric at 15/30/60 deg and multiple azimuths.

BASE, SEALS, CONDENSATE DRAINAGE AND ROOF INTERFACE

Exploded assembly and controlled interface dimensions; curb and anchorage remain project-specific

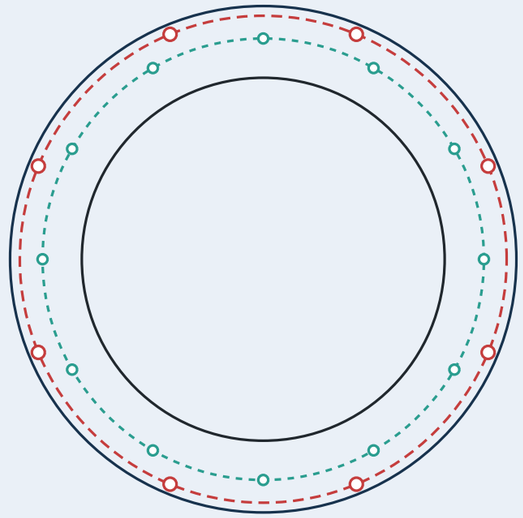
UNITS: mm unless noted | SCALE: diagrammatic

EXPLODED SECTION - ASSEMBLY ORDER



Assembly: gasket surfaces flat within 0.5; tighten M6 fasteners gradually in a star pattern only to gasket compression stop.

BASE PLATE B-01 - TOP VIEW AND HOLE CONTROL



ROOF HOLES: HOLD

Do not drill PCD750 curb holes until curb, anchor edge distance, wind uplift and flashing are approved.

DETAIL S-1 - DOME CLAMP AND THERMAL MOVEMENT



Allow radial sliding; do not rigidly pin PC

BOM - COLLECTOR HEAD

ITEM	QTY	DESCRIPTION
D-01	1	Formed UV PC dome, 8 start / 3.2 min
C-01	1	304 SS clamp ring OD715/ID652 x 4
S-01	1	Closed-cell EPDM OD710/ID650 x 6
S-02	1	EPDM O-ring cord DIA6, 60A
B-01	1	304 SS base OD780/aperture559 x 3
G-01	1	SS gutter OD640/ID560 x 18 deep
O-01	1	Specular Al cone 600/400 x H173.2
H-01	12	M6 x 25 316 SS + bonded EPDM isolation
H-02	6	M4 SS reflector posts, H173 nominal

SHEET 4 EXPLAINED - BASE, SEALS, DRAINAGE

Assembly stack, thermal movement, and the roof-hole HOLD

ASSEMBLY ORDER

1. **D-01 dome**
2. **C-01 304 SS clamp ring**
3. **S-01 EPDM flange gasket**
4. **B-01 304 SS base**
5. **S-02 EPDM O-ring**
6. **G-01 annular gutter + 4 exterior drains**
7. **O-01 redirector on six M4 posts**
8. **Project curb / flashing / anchors - TBD**

FASTENER METHOD

Keep gasket surfaces flat within 0.5 mm. Tighten M6 fasteners gradually in a star pattern only to the gasket-compression stop.

SEAL + THERMAL MOVEMENT

- C-01 clamp ring thickness: 4 mm.
- D-01 dome flange: 8 mm nominal before forming effects at cap.
- S-01 EPDM t6 is compressed to 4.2-4.8 mm.
- B-01 base thickness: 3 mm.
- Allow radial sliding. Do not rigidly pin the polycarbonate flange.
- The twelve radial 8 x 18 slots on PCD680 accommodate about 1.9 mm radial growth for Delta-T = 80 C.

DRAINAGE PATH

- G-01 is an annular gutter, OD640 / ID560 x 18 deep.
- Provide four DIA8 drains to the exterior above flashing.
- Never drain into the reflective tube.
- During dry assembly, verify drainage fall and that all four drains remain clear.

BASE HOLE CONTROL

- Dome-clamp pattern: 12 x DIA7 on PCD680.
- Roof/curb pattern: 8 x DIA9 on PCD750 is TBD.
- Do not drill PCD750 holes until the project curb, anchor edge distances, uplift load, and flashing details are approved.

FABRICATION, ASSEMBLY, TEST AND RELEASE CHECKLIST

Prototype workflow and evidence required before any permanent roof installation

A. MANUFACTURE AND ASSEMBLE THE COLLECTOR HEAD

01	**PROCURE**	UV-stabilized optical PC 8.0; 304 SS plate/ring; 95%+ specular Al; compatible EPDM.
02	**FORM DOME**	Pre-dry, heat, free-blow through DIA650 fixture to 300 height; hold to rigid; retain process record.
03	**INSPECT DOME**	50 mm grid thickness map; reject below 3.2; visual check; trim OD720 and machine twelve 8 x 18 slots.
04	**MAKE REFLECTOR**	Cut R600/R400 half-annulus, 180 deg + 15 lap; roll to DIA600/DIA400 x H173.2.
05	**MAKE METALWORK**	Kit base OD780/aperture559, clamp ring OD715/ID652, support rings and gutter; deburr/passivate.
06	**DRY ASSEMBLE**	Install reflector and gutter. Check DIA400 clear path, drainage fall, runout and tube alignment.
07	**SEAL DOME**	Fit O-ring and EPDM gasket; place dome; install isolated M6 fasteners; tighten in star sequence.
08	**BENCH TEST**	Water, air, pressure, thermal cycle and optical tests. Record results and corrective actions.
09	**ROOF DESIGN**	Obtain roof/site loads and code. Engineer curb, anchors, insulation and flashing for the project.
10	**RELEASE**	Issue project-specific stamped/approved installation set only after structural and weather tests pass.

B. VERIFICATION MATRIX - DOCUMENT EVERY RESULT

TEST / INSPECTION	PROTOTYPE ACCEPTANCE	STATUS
Dome dimensions	OD720; cap DIA650; H300 +/- 3	OPEN
Dome thickness	50 grid; every point >= 3.2	OPEN
Visual / optical	No haze, webbing, bubbles or deep scratches	OPEN
Drainage	All 4 drains clear; no water enters tube	OPEN
Air leakage	ASTM E283 / adopted NAFS criterion	OPEN
Water penetration	ASTM E331 / adopted NAFS criterion	OPEN
Structural pressure	ASTM E330 at project design pressures	HOLD
Impact / hail	Classification required by adopted code	HOLD
Fire / roof covering	Classification required by adopted code	HOLD
Thermal cycling	No cracks, seal loss or fastener binding	OPEN
UV exposure	Supplier qualification + project requirement	OPEN
Photometric	Lux maps at 15/30/60 deg, several azimuths	OPEN
Roof curb / anchors	Engineered for roof and site loads	HOLD

C. PROJECT-SPECIFIC RELEASE DATA - ALL FIELDS REQUIRED

Project / address TBD

Design wind pressure + / - TBD

Roof material / assembly TBD

Ground / roof snow load TBD

Roof slope and curb height TBD

Hail and fire classification TBD

Adopted building code / edition TBD

Engineer / approval reference TBD

NO BUILDING INSTALLATION RELEASE UNTIL ALL FIELDS AND HOLD TESTS ARE CLOSED

D. TECHNICAL BASIS AND LIMITS

[1] POLYVANTIS, LEXAN Sheet Processing, Forming and Fabricating Guide, 2025 - drying, dome-forming and thermoforming guidance.

[2] POLYVANTIS, LEXAN XL102UV Sheet Datasheet - UV protection, transmission, thermal and forming properties.

[3] Public VELUX tubular unit skylight specification - nominal 559 mm unit and 3.18 mm minimum glazing reference.

[4] Alanod daylight-system materials - enhanced-specular aluminum reflectance capability up to about 98%.

[5] FGIA NAFS overview - tubular daylight device product classification and performance testing framework.

[6] Exact material grade, adopted code, NAFS edition, curb, flashing, anchors and load ratings remain project-specific.

polyvantis.com - LEXAN Sheet Processing, Forming and Fabricating Guide 2025

polyvantis.com - LEXAN XL102UV Sheet Datasheet, Americas

commercial.veluxusa.com/resources/tubular | alanod.com/applications/lighting-industry/daylight-systems

fgiaonline.org/resources/technical-resources/nafs-overview

PROTOTYPE TEST + RELEASE CHECKLIST

Evidence that must be collected before any permanent installation package

A. FABRICATION + TEST FLOW

1. **Procure**
   UV optical PC 8.0; 304 SS; >=95% specular Al; compatible EPDM.
2. **Form dome**
   Pre-dry, heat, free-blow through DIA650 to 300 height; retain process record.
3. **Inspect dome**
   50 mm grid thickness map; visual inspection; trim OD720; machine slots.
4. **Make reflector**
   R600/R400 half-annulus, 180 deg + 15 lap; roll to 600/400 x H173.2.
5. **Make metalwork**
   Base, clamp ring, support rings and gutter; deburr/passivate.
6. **Dry assemble**
   Check DIA400 clear path, drainage fall, runout, tube alignment.
7. **Seal dome**
   Fit O-ring + EPDM; isolated M6 hardware; star tightening.
8. **Bench test**
   Water, air, pressure, thermal cycle and optical tests; record corrective actions.
9. **Roof design**
   Obtain roof/site loads and code; engineer curb, anchors, insulation, flashing.
10. **Release**
    Issue project-specific approved/stamped installation set only after required tests pass.

B. VERIFICATION MATRIX - STATUS

OPEN - document result

- Dome dimensions
- Dome thickness map
- Visual / optical
- Drainage
- Air leakage
- Water penetration
- Thermal cycling
- UV exposure
- Photometric testing

HOLD - project input / approval

- Structural pressure at project design pressures
- Impact / hail classification
- Fire / roof-covering classification
- Roof curb / anchors

C. PROJECT RELEASE DATA - ALL REQUIRED

- Project / address
- Roof material / assembly
- Roof slope + curb height
- Adopted building code / edition
- Design wind pressure + / -
- Ground / roof snow load
- Hail + fire classification
- Engineer / approval reference

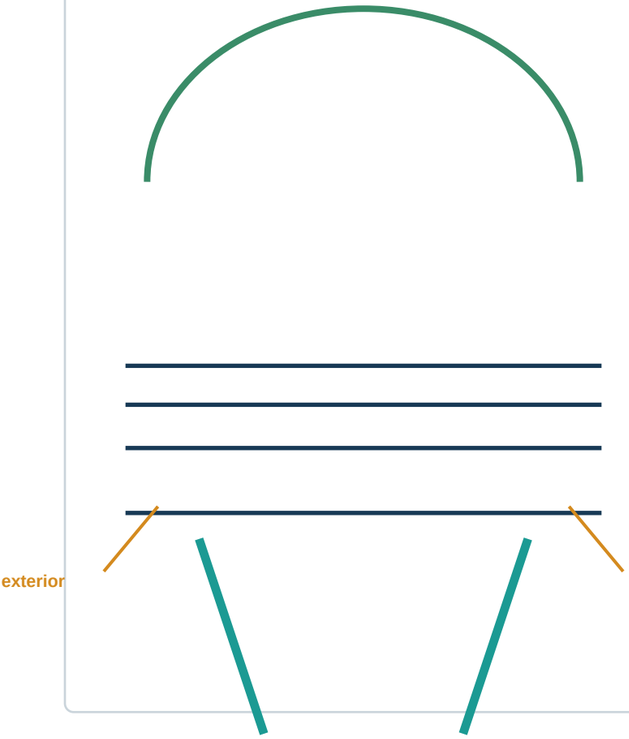
NO INSTALLATION RELEASE

Do not issue a permanent building-installation release until all project fields and HOLD tests are closed.

COLLECTOR TERMINOLOGY, COMPONENTS & INTERFACES

A clearer key to the collector stack and the fixed prototype geometry

A. COMPONENT STACK - WHAT EACH PART DOES



- D-01 UV polycarbonate dome - captures diffuse sky light and direct sun; formed from 8.0 mm starting sheet with 3.2 mm minimum formed wall.
- C-01 304 SS clamp ring - compresses the dome gasket through isolated M6 hardware.
- S-01 closed-cell EPDM gasket - 6 mm nominal, compressed to 4.2-4.8 mm in the clamp detail.
- S-02 EPDM O-ring - secondary seal; groove is supplier-sized.
- B-01 304 SS base - OD780 with DIA559 tube aperture; roof-hole pattern is not released.
- G-01 annular gutter - four drains route condensate to the exterior, never into the reflective tube.
- O-01 specular aluminum redirector - DIA600/DIA400 x H173.2, 60 deg mirror angle.

C-01 SS clamp ring

S-01 EPDM gasket

B-01 SS base

G-01 condensate gutter

O-01 optical redirector

B. FIXED PROTOTYPE GEOMETRY - DO NOT CONFUSE WITH ROOF RELEASE

FEATURE	PROTOTYPE VALUE
Tube clear diameter	559 mm
Dome forming aperture	650 mm
Dome height	300 +/- 3 mm
Dome flange OD	720 mm
Dome minimum formed wall	3.2 mm everywhere
Collector base	304 SS, OD780 x 3 mm; aperture DIA559
Redirector	DIA600 / DIA400 x H173.2; beta 60 deg
Seal concept	6 mm EPDM gasket + secondary 6 mm O-ring
Roof interface	PCD750 provision only - NOT RELEASED

IMPORTANT DISTINCTION

The collector head geometry can be prototyped, but curb, flashing, anchors and structure cannot be released from this package because roof system, slope, project location, adopted code, wind pressure and snow load remain project inputs.

C. OPTICAL PATH IN PLAIN LANGUAGE

- The clear dome accepts diffuse sky light and direct sunlight over a wide azimuth range.
- The annular 60 deg redirector is set so a 30 deg solar-altitude ray is approximately redirected vertically.
- The central DIA400 opening allows high-angle sunlight to pass directly toward the tube.
- Optical output is still a design intent: the source requires lux mapping at 15, 30 and 60 deg incidence and multiple azimuths.

DOME FORMING - DETAILED PROCESS & QA RECORD

The source forming sequence expanded so operators and reviewers can see every controlled value

A. FORMING SEQUENCE

1. **START MATERIAL**
   UV-stabilized optical PC, 8.0 mm starting thickness; keep protected.
2. **PRE-DRY**
   125 +/- 3 C for 36 h; follow sheet supplier.
3. **HEAT**
   Suggested forming trial window 170-180 C.
4. **FREE-BLOW**
   Clamp air-tight over DIA650 aperture; use dry regulated air, gauge and relief.
5. **CONTROL HEIGHT**
   Stop at 300 +/- 3 mm using calibrated non-contact gauge.
6. **HOLD / RELEASE**
   Hold shape until rigid; source notes about 120-125 C part temperature.
7. **THICKNESS MAP**
   Mark 50 x 50 inspection grid; map thickness over the cap.
8. **TRIM + SLOT**
   Trim flange to DIA720 concentric within 1.0 TIR; machine twelve 8 x 18 radial slots on PCD680.
9. **VISUAL QA**
   Inspect haze, bubbles, webbing, scratches and UV-side reversal.
10. **RECORD**
    Retain sheet lot, heat cycle, height, pressure history and thickness map.

B. WHY 8 mm STARTING SHEET IS USED

Cap area = 614,500 mm2

Aperture area = 331,800 mm2

Draw ratio = 1.852

8.0 / 1.852 = 4.32 mm average

The source calculation explains the material budget only. Local thinning still controls acceptance, and every mapped cap reading must be at least 3.2 mm.

REJECT CONDITION

Any mapped dome thickness below 3.2 mm is a rejection, even if the calculated average is above 3.2 mm.

C. BLANK THICKNESS-MAP RECORD - ENTER ACTUAL MEASUREMENTS

Use the project/shop record to map the 50 x 50 grid over the actual cap. This blank is a documentation aid, not a replacement for the source geometry.

Minimum reading: _____ mm

Inspector / date: _____

Sheet lot: _____

Result: PASS / REJECT

OPTICAL REDIRECTOR - GEOMETRY & RAY LOGIC

A more readable explanation of why the cone is 60 degrees and what the acceptance test still needs to prove

A. FINISHED O-01 GEOMETRY

FEATURE	SOURCE VALUE
Top diameter	600 mm
Bottom / clear diameter	400 mm
Radial change	100 mm
Slant length	200.0 mm
Axial height	173.2 mm
Mirror angle beta	60 deg to horizontal
Material	Enhanced-specular Al, 0.8 mm; visible R >=95%
Support	OD630 / ID600 ring; six M4 posts
Finished tolerance	DIA600 / DIA400 / H173.2 each +/-1.0 mm
Runout	1.5 mm max at support ring

B. RAY RELATION - TARGETED AT 30 DEG SOLAR ALTITUDE

beta = 45 deg + alpha0 / 2

beta = 45 + 30/2 = 60 deg

For the meridional 2D design, alpha0 = 30 deg is the reference condition that gives approximately vertical reflected output.

alpha = 15 deg delta = -15 deg

alpha = 30 deg delta = 0 deg

LIMIT OF THE SIMPLE RAY MODEL

The source explicitly says full 3D performance varies with solar azimuth, dome refraction, surface errors and tube losses. The equation sets cone slope; it does not release optical performance.

C. FLAT PATTERN + FABRICATION CONTROLS

PATTERN	JOIN / SURFACE	FINAL CHECK
Half-annular sector	Outer development radius 600 mm; inner 400 mm; included angle 180 deg; add 15 mm lap.	Verify a paper template before cutting optical aluminum.
Seam	Roll with film on optical face; seam at north/reference side; flush rivet or bonded lap.	No fastener head may project into mirror face or DIA400 clear opening.
Surface	Soft covered rolls and gloves; keep inner face smooth and clean.	Reflectance certificate 95% visible or better; clean only by approved non-abrasive method.
Photometric acceptance	Lux map at diffuser for 15, 30 and 60 deg elevation and multiple azimuths.	This is the actual optical acceptance, not the 2D ray sketch alone.

SEALING, THERMAL MOVEMENT & DRAINAGE

Assembly logic that prevents the seal system from fighting thermal expansion or trapping condensate

A. ASSEMBLY STACK - TOP TO BOTTOM

1	**D-01 dome**	UV PC; flange t8 nominal
2	**C-01 clamp ring**	304 SS, OD715 / ID652 x 4
3	**S-01 gasket**	Closed-cell EPDM, t6
4	**B-01 base**	304 SS, OD780 / aperture DIA559 x 3
5	**S-02 O-ring**	EPDM DIA6; supplier-sized groove
6	**G-01 gutter**	OD640 / ID560 x 18 deep; four exterior drains
7	**O-01 redirector**	Six M4 support posts
8	**Project curb**	Height, flashing and anchors TBD

B. CRITICAL SEAL / MOVEMENT RULES

- Gasket surfaces are flat within 0.5 mm before clamping.
- Tighten M6 fasteners gradually in a star pattern only to the gasket compression stop.
- S-01 EPDM t6 is compressed to 4.2-4.8 mm in the source clamp detail.
- Allow radial sliding of the polycarbonate flange; do not rigidly pin the PC.
- The twelve 8 x 18 radial slots on PCD680 accommodate about 1.9 mm radial growth for Delta-T = 80 C.
- All four gutter drains remain clear and discharge to the exterior above flashing - never into the reflective tube.

ROOF HOLES REMAIN ON HOLD

Do not drill the PCD750 curb holes until curb geometry, anchor edge distance, wind uplift and flashing are approved.

C. BEFORE / DURING / AFTER CLAMPING

BEFORE CLAMPING	DURING CLAMPING	AFTER CLAMPING
Confirm flange, clamp ring, gasket and base are clean and aligned.	Use isolated M6 hardware and tighten gradually in a star pattern.	Verify gasket compression is within the source range and the dome is not rigidly pinned.
Confirm O-ring/groove concept is compatible with selected supplier detail.	Stop at gasket compression stop - the source does not call for rigid crushing of the joint.	Check all four drains are open and no water path leads into DIA559 tube.
Confirm PCD750 roof holes are still withheld unless project roof design is approved.	Maintain the intended radial sliding allowance through the slotted dome flange.	Retain access for water, air, pressure, thermal-cycle and optical bench testing.

COLLECTOR TEST EVIDENCE MATRIX

The source verification matrix reorganized by what is OPEN now versus what remains project-dependent

A. OPEN - DOCUMENT THE PROTOTYPE RESULT

TEST / INSPECTION	PROTOTYPE ACCEPTANCE	RESULT / RECORD REF.
Dome dimensions	OD720; cap DIA650; H300 +/-3	
Dome thickness	50 mm grid; every point >=3.2	
Visual / optical	No haze, webbing, bubbles or deep scratches	
Drainage	All 4 drains clear; no water enters tube	
Air leakage	ASTM E283 / adopted NAFS criterion	
Water penetration	ASTM E331 / adopted NAFS criterion	
Thermal cycling	No cracks, seal loss or fastener binding	
UV exposure	Supplier qualification + project requirement	
Photometric	Lux maps at 15/30/60 deg, several azimuths	

B. HOLD - PROJECT CRITERION OR APPROVAL NEEDED

HOLD ITEM	SOURCE ACCEPTANCE	WHAT IS STILL MISSING
Structural pressure	ASTM E330 at project design pressures	Project design pressure required
Impact / hail	Classification required by adopted code	Adopted code + classification required
Fire / roof covering	Classification required by adopted code	Adopted code + roof system required
Roof curb / anchors	Engineered for roof and site loads	Roof assembly, loads and engineering required

STATUS MEANING

OPEN does not mean automatically passed; it means the prototype result still has to be documented. HOLD means the package cannot define the final acceptance without project-specific information or approval.

C. TEST RECORD HEADER - USE WITH EACH BENCH / PROJECT TEST

Test / inspection _____ **Date / operator** _____

Prototype serial / part lot _____ **Procedure / standard reference** _____

Test setup / condition _____ **Measured result** _____

Pass / fail / HOLD _____ **Corrective action (if any)** _____

Retest reference _____ **Reviewer / approval** _____

PROJECT RELEASE GATE & HANDOFF WORKSHEET

A final decision page that keeps prototype fabrication separate from permanent roof installation approval

A. ALL PROJECT-SPECIFIC RELEASE DATA REQUIRED

Project / address	REQUIRED
Roof material / assembly	REQUIRED
Roof slope and curb height	REQUIRED
Adopted building code / edition	REQUIRED
Design wind pressure + / -	REQUIRED
Ground / roof snow load	REQUIRED
Hail and fire classification	REQUIRED
Engineer / approval reference	REQUIRED

B. RELEASE DECISION

1	Prototype fabrication complete?	NO -> HOLD / NO INSTALLATION RELEASE
2	OPEN tests documented?	NO -> HOLD / NO INSTALLATION RELEASE
3	Roof / site / code data complete?	NO -> HOLD / NO INSTALLATION RELEASE
4	HOLD tests and engineering approvals closed?	NO -> HOLD / NO INSTALLATION RELEASE

SOURCE RELEASE RULE

No permanent building installation release until all project fields and HOLD tests are closed.

C. HANDOFF / APPROVAL RECORD

Prototype fabrication record	Dome thickness map	Water / air test records
Structural pressure test / approval	Impact / hail classification	Fire / roof-covering classification
Photometric test record	Roof curb + anchor engineering	Flashing / roof interface approval
Approved installation set reference	Responsible engineer / reviewer	Release date